

Task Questions

1. Question 1

Answer:

Amazon EFS is a fully managed service that provides file system storage for Linux-based workloads. Multiple EC2 instances can be accessed. It also allows for a common interface, like a shared workspace or data source for workloads and application that run on more than one Amazon EC2 instance. In addition, it can also scale automatically from GBs to PBs of data without needing to provision storage.

Amazon EC2 instance store and Amazon EBS provide good choices for storing both root and data volume for a single instance. This is a block store. EBS volumes are independent storage units that can be attached to EC2 instances and used as primary storage for operating systems, databases, and other applications. So, the main difference between them is that EBS volumes are accessible to only one EC2 instance at a time unlike EFS. This property makes it suitable for instances that require low-latency access to block storage.

2. Question 2

Answer:

Amazon EFS uses the Network File System (NFS) version 4.x protocol. It can be used with any AMIs that support this protocol.

3. Question 3

Answer:

It is called **Amazon FSx (File System)**, it provides fully managed Windows file servers, backed by a fully-native Windows file system.

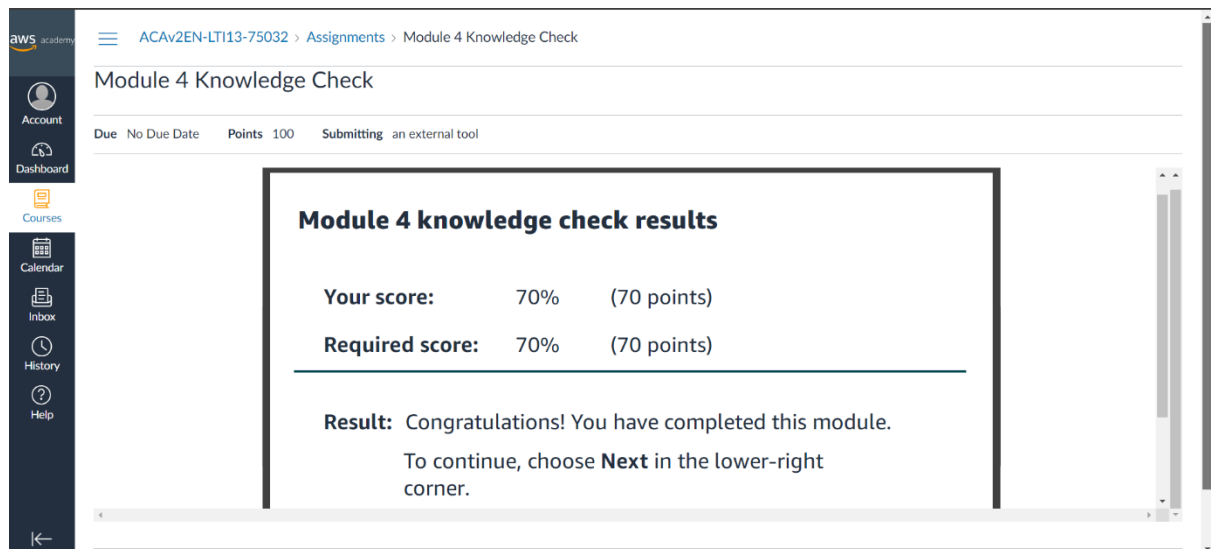
It uses the following protocols:

- New Technology File System (NTFS).
- Server Message Block (SMB).
- Distributed File System (DFS).
- Microsoft Active Directory, service that administers and organizes enterprise resources (including users, groups, and network resources).

For a data volume that serves multiple Microsoft Windows instances, use Amazon FSx for Windows File Server.

Quiz

1. Knowledge Check for Module 4



This knowledge check was comprised of the most important questions throughout the module 4. The questions asked was kind of a summary to the whole module 4. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

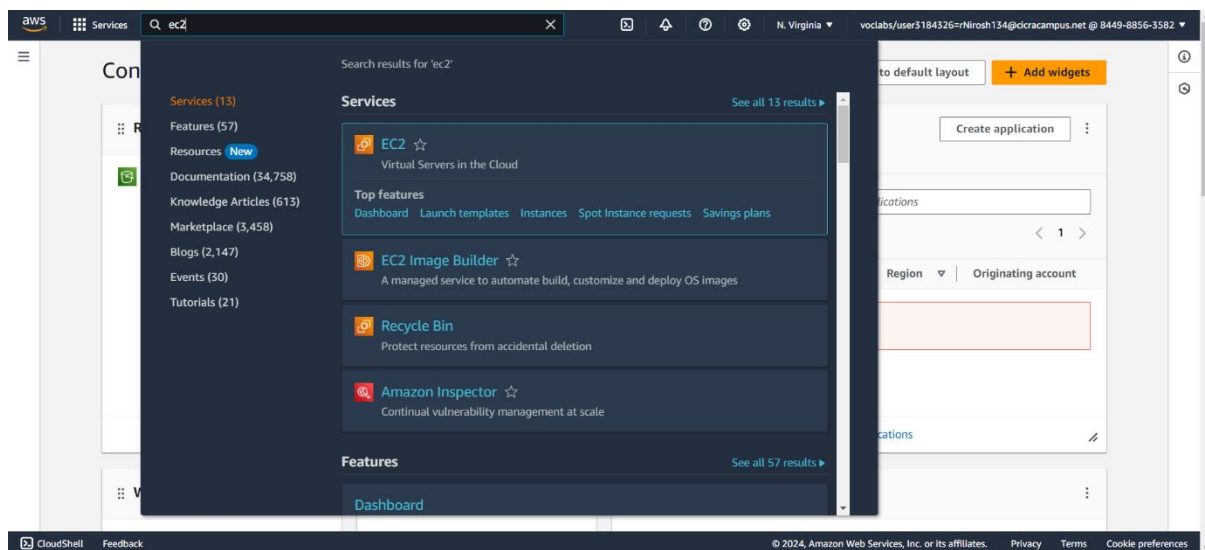
It was mostly about the EFS and EBS. What they are used for, their speed, their benefits and drawbacks, like the autoscaling and high availability are two of them. The security features they have. The file systems are managed for Linux by EFS whereas for Windows, Amazon FSx is used to manage the file systems. Then the pricing options for EC2 are asked as questions, they usually depend on the on-demand instances and spot instances. This is some of the

Module 4 Guided Lab - Introducing Amazon Elastic File System (Amazon EFS)

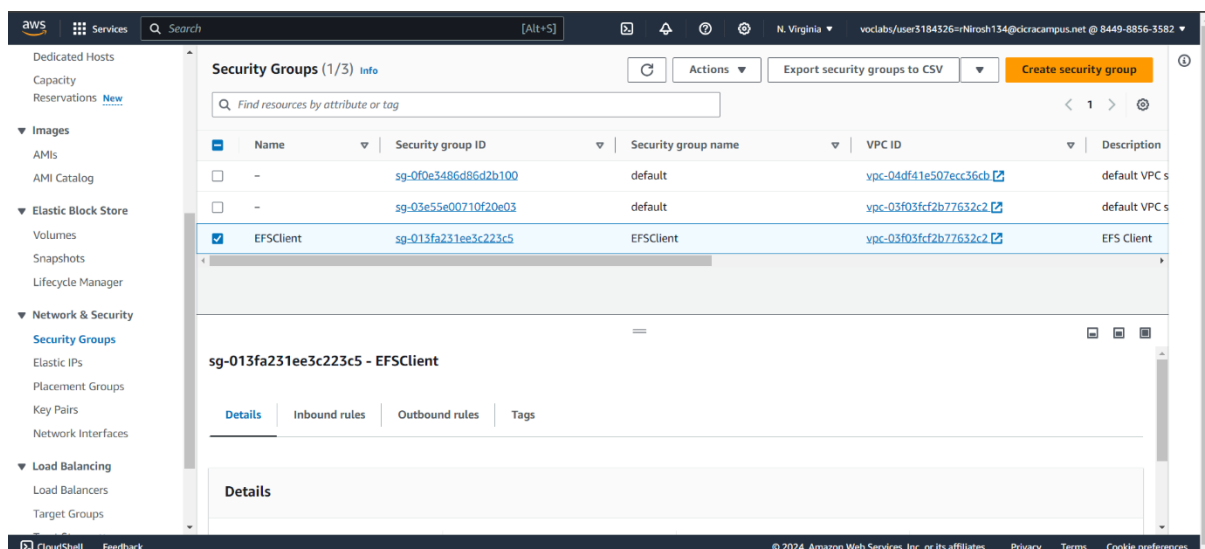
Task 1: Creating a security group to access your EFS file system

In this task I'll be setting up a security group in AWS to regulate inbound access on TCP port 2049 for NFS which is crucial for managing EFS mount targets.

First, I started by accessing the AWS Management Console and navigating to the EC2 service, where we can manage security groups.



Within the EC2 service, we navigate to the Security Groups section. Security groups act as virtual firewalls that control the traffic for the instances.



Identify an existing security group called "EFSClnt" and copy its Security Group ID.

Created a new security group named "EFS Mount Target" specifically to manage inbound NFS access from EFS clients.

The screenshot shows the 'Create security group' page in the AWS Management Console. The page is titled 'Create security group' with a sub-header 'Basic details'. The 'Security group name' field is filled with 'EFS Mount Target'. The 'Description' field is filled with 'Inbound NFS access from EFS clients'. The 'VPC' dropdown menu is set to 'vpc-03f03fcf2b77632c2 (Lab VPC)'. Below the 'Basic details' section, there is an 'Inbound rules' section which is currently empty, indicating that no inbound rules have been added yet.

Then name and description given for the security group and specifying VPC.

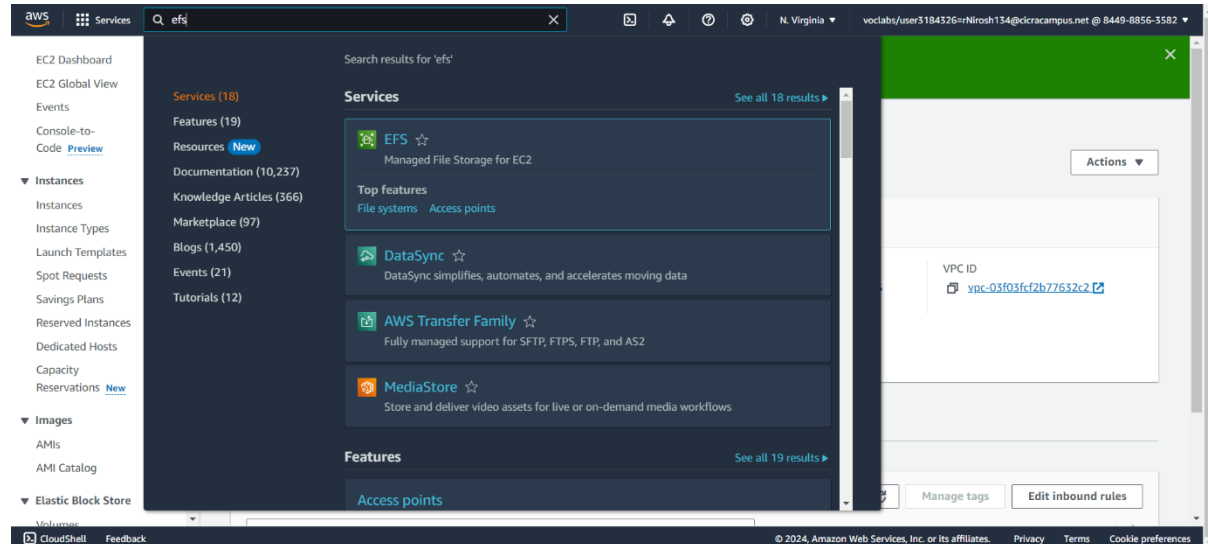
An inbound rule is set up within the new security group to allow NFS traffic. The source will be specified as the "EFSClnt" security group

The screenshot shows the 'Inbound rules' section of the 'EFS Mount Target' security group. The 'VPC' dropdown menu is set to 'vpc-03f03fcf2b77632c2 (Lab VPC)'. The 'Inbound rules' section is active, showing a table with columns: Type, Protocol, Port range, Source, and Description - optional. A single rule is added with the following details: Type: NFS, Protocol: TCP, Port range: 2049, Source: Custom (sg-013fa231ee3c223c5), and Description - optional: (empty). The 'Add rule' button is visible below the table. The 'Outbound rules' section is also visible, showing a table with columns: Type, Protocol, Port range, Destination, and Description - optional. A single rule is added with the following details: Type: All traffic, Protocol: All, Port range: All, Destination: Custom (0.0.0.0/0), and Description - optional: (empty).

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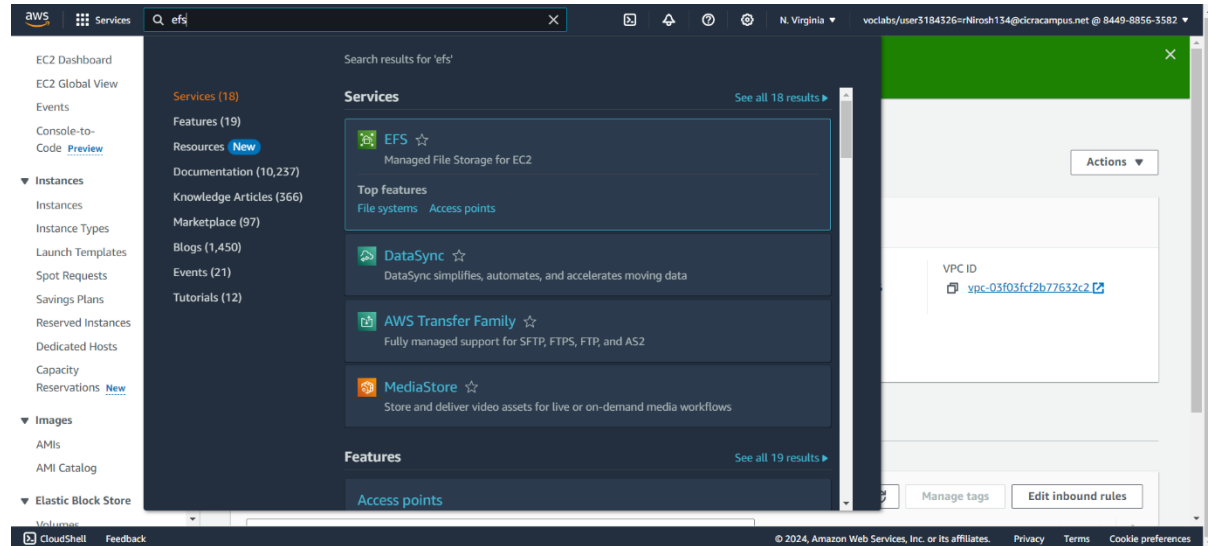
Name: Nirosh Ravindran
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Finally, a security group is set up with the specified the configuration.

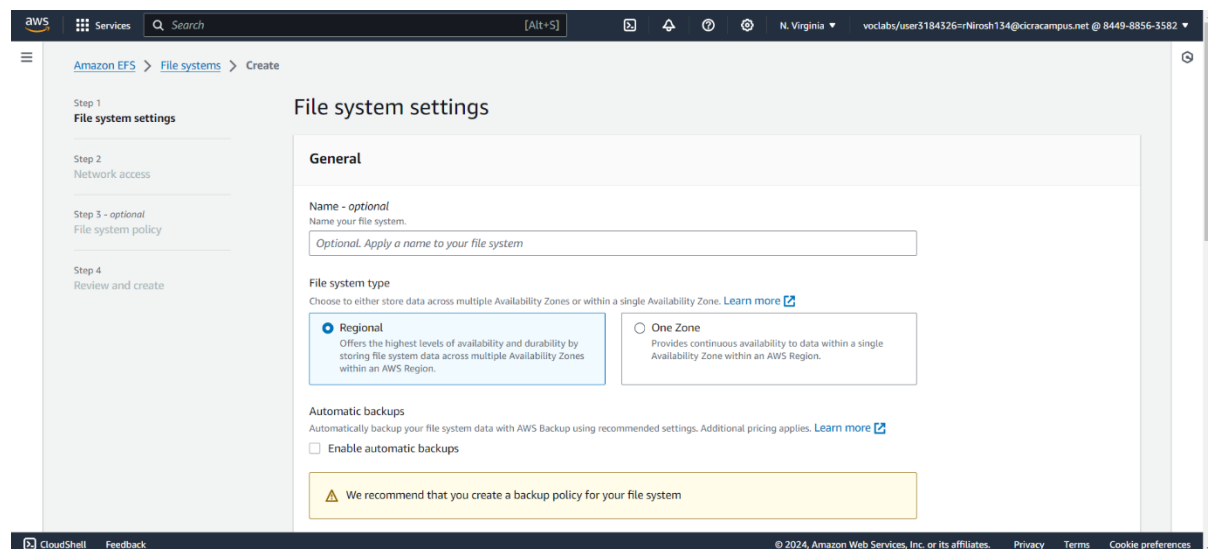


Task 2: Creating an EFS file system

I'm setting up an Elastic File System (EFS) in AWS. I create a new file system, customize its settings, specify the VPC where it will be used, configure security groups for mount targets, review the configuration, and create the file system. Finally, I confirm its creation and wait for it to become available, ensuring secure access for EC2 instances within my chosen VPC.



I navigate to the EFS service in the AWS Management Console and select "Create file system" to initiate setting up a new file system.



I choose "Customize" to specify detailed settings. In Step 1, I uncheck automatic backups and select "None" for lifecycle management.

Regional
Offers the highest levels of availability and durability by storing file system data across multiple Availability Zones within an AWS Region.

One Zone
Provides continuous availability to data within a single Availability Zone within an AWS Region.

Automatic backups
Automatically backup your file system data with AWS Backup using recommended settings. Additional pricing applies. [Learn more](#)

☐ Enable automatic backups

⚠️ We recommend that you create a backup policy for your file system

Lifecycle management
Automatically save money as access patterns change by moving files into the Infrequent Access (IA) or Archive storage class. [Learn more](#)

Transition into Infrequent Access (IA)
Transition files to IA based on the time since they were last accessed in Standard storage.
None

Transition into Archive
Transition files to Archive based on the time since they were last accessed in Standard storage.
90 day(s) since last access

Transition into Standard
Transition files back to Standard storage based on when they are first accessed in IA or Archive storage.
None

Encryption
Choose to enable encryption of your file system's data at rest. Uses the AWS KMS service key (aws/elasticfilesystem) by default. [Learn more](#)

☒ Enable encryption of data at rest

[Customize encryption settings](#)

I also tag the file system, naming it "My First EFS File System".

Elastic (Recommended)
Use this mode for workloads with unpredictable I/O. With Elastic Throughput, performance automatically scales with your workload activity and you only pay for the throughput you use (data transferred for your file systems per month). [Learn more](#)

Provisioned
Use this mode if you can estimate your workload's throughput requirements. With Provisioned mode, you configure your file system's throughput and pay for throughput provisioned.

Additional settings

Tags optional

Add tags to associate key-value pairs to your resource. [Learn more](#)

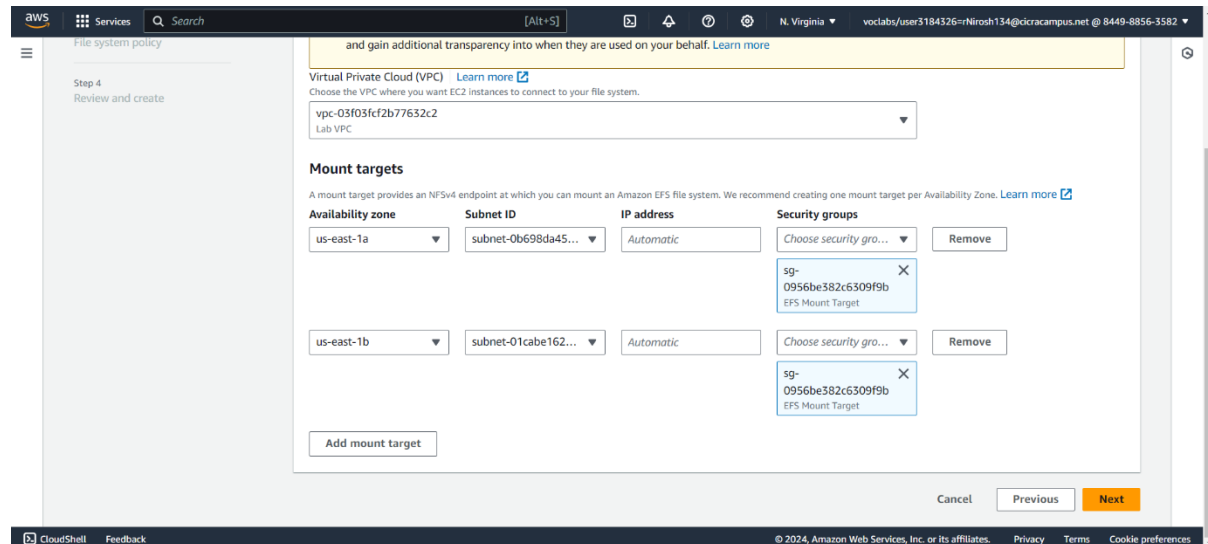
Tag key	Tag value - optional	
Name	My First EFS File System	Remove tag

[Add tag](#)

You can add 49 more tag(s)

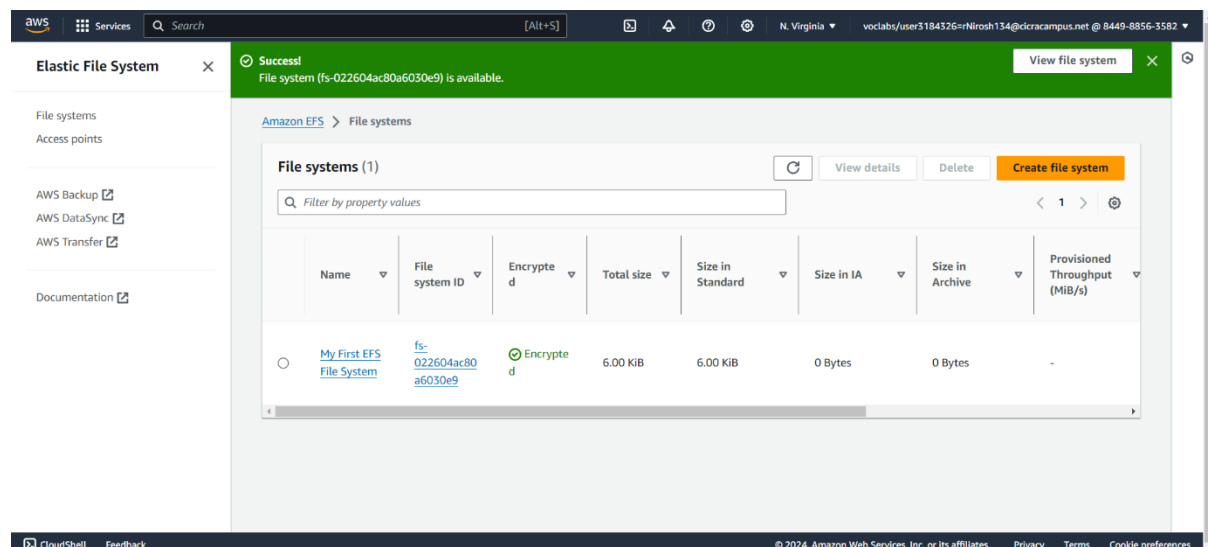
[Cancel](#) [Next](#)

Selected the desired VPC "Lab VPC". Then removed the default security group from each Availability Zone's mount target and attach the "EFS Mount Target" security group to each one.



I confirm the configuration and choose "Create" to create the file system.

Shortly after creation, the file system state changes to "Available", followed by the mount targets' state a few minutes later. I wait for the mount target states to change to "Available" before proceeding further.

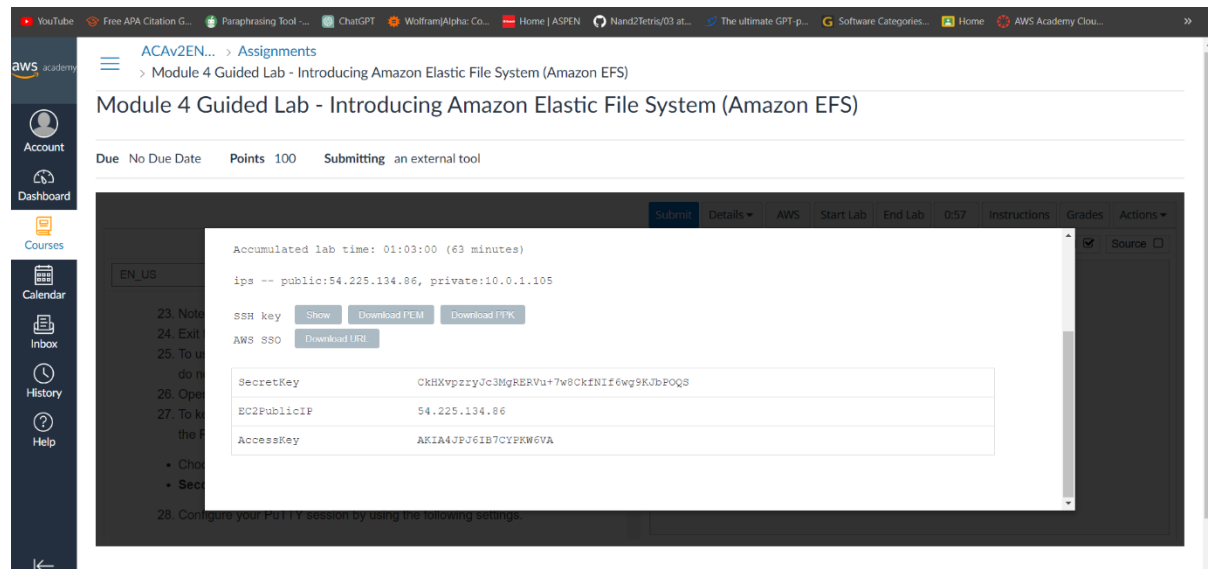


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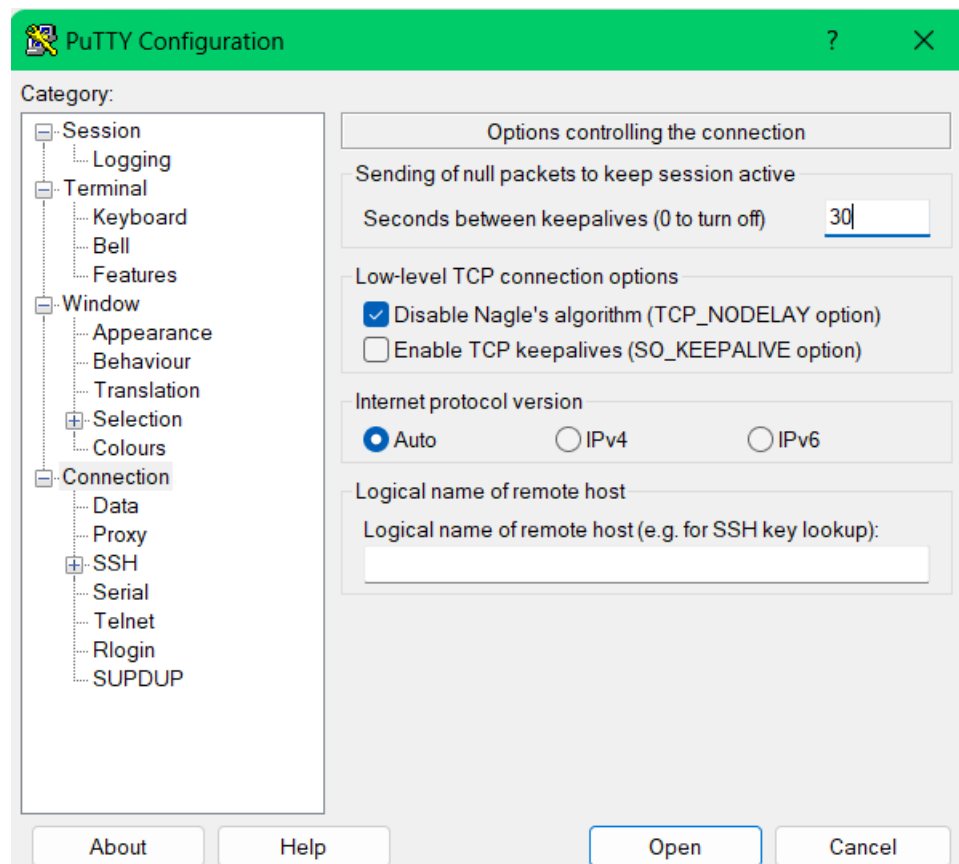
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Task 3: Connecting to your EC2 instance via SSH

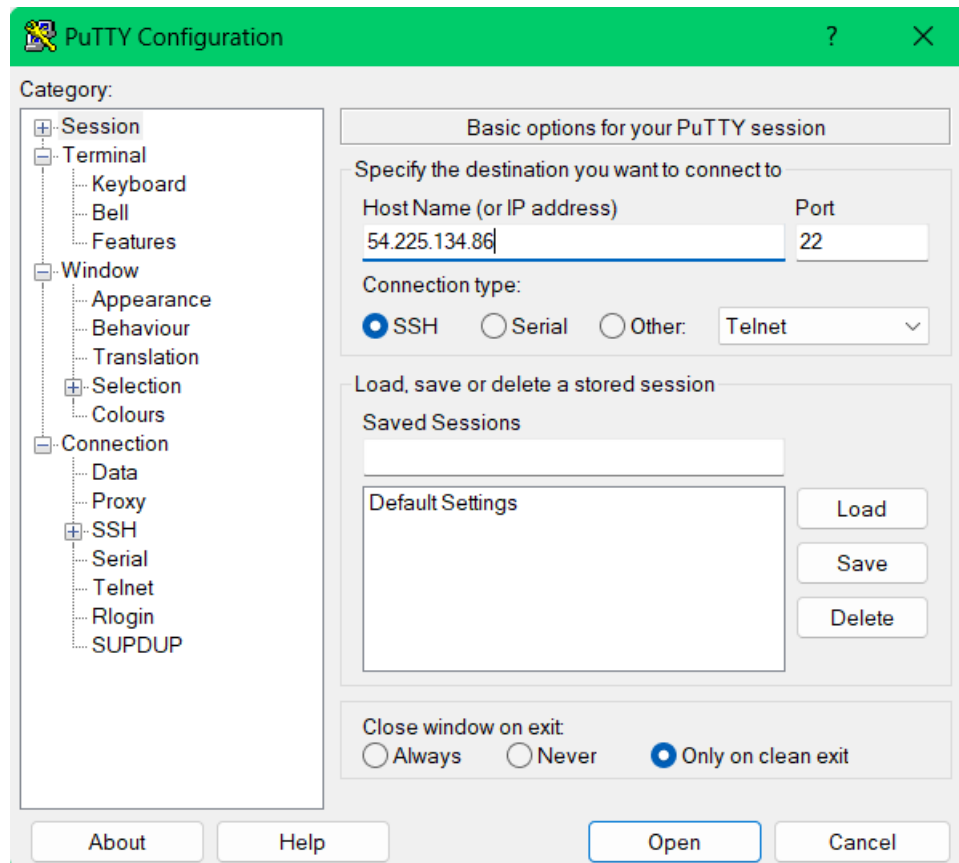
Opened the Details dropdown menu and select "Show" to reveal a Credentials window. Chose the "Download PPK" button and save the labsuser.ppk file to my computer. Then saved the EC2PublicIP address if it's displayed.



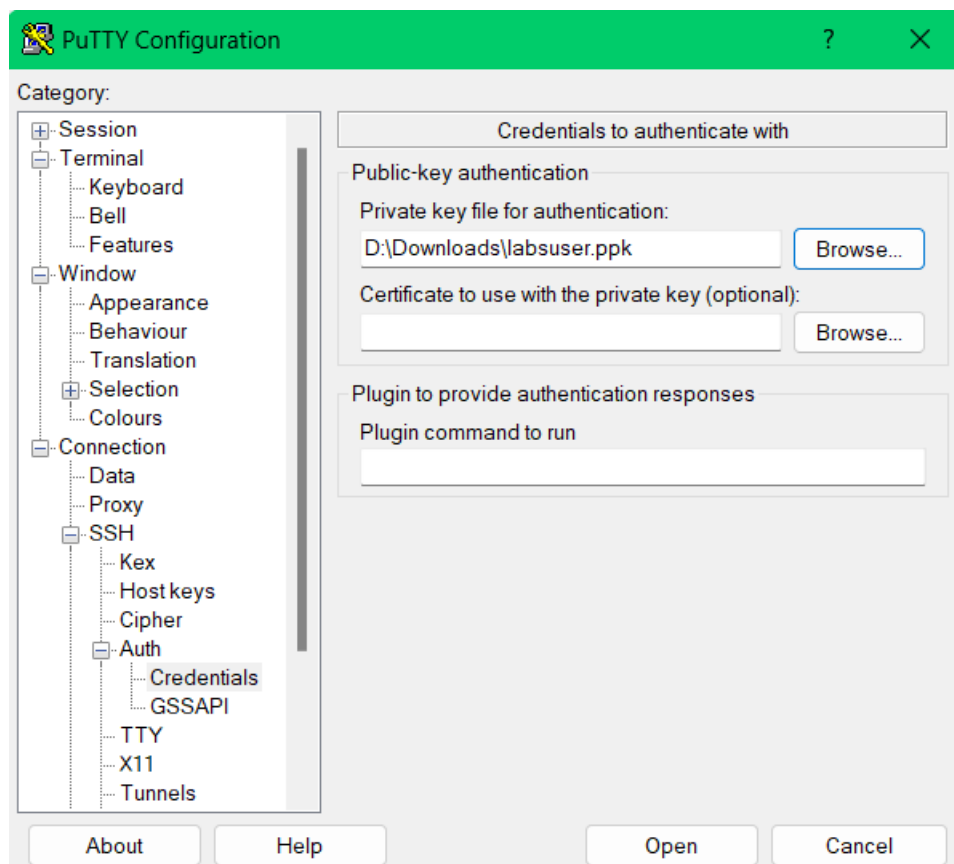
Opened the putty.exe, a terminal emulator application and configured PuTTY to keep the SSH session open for a longer period by setting the timeout to 30 seconds.



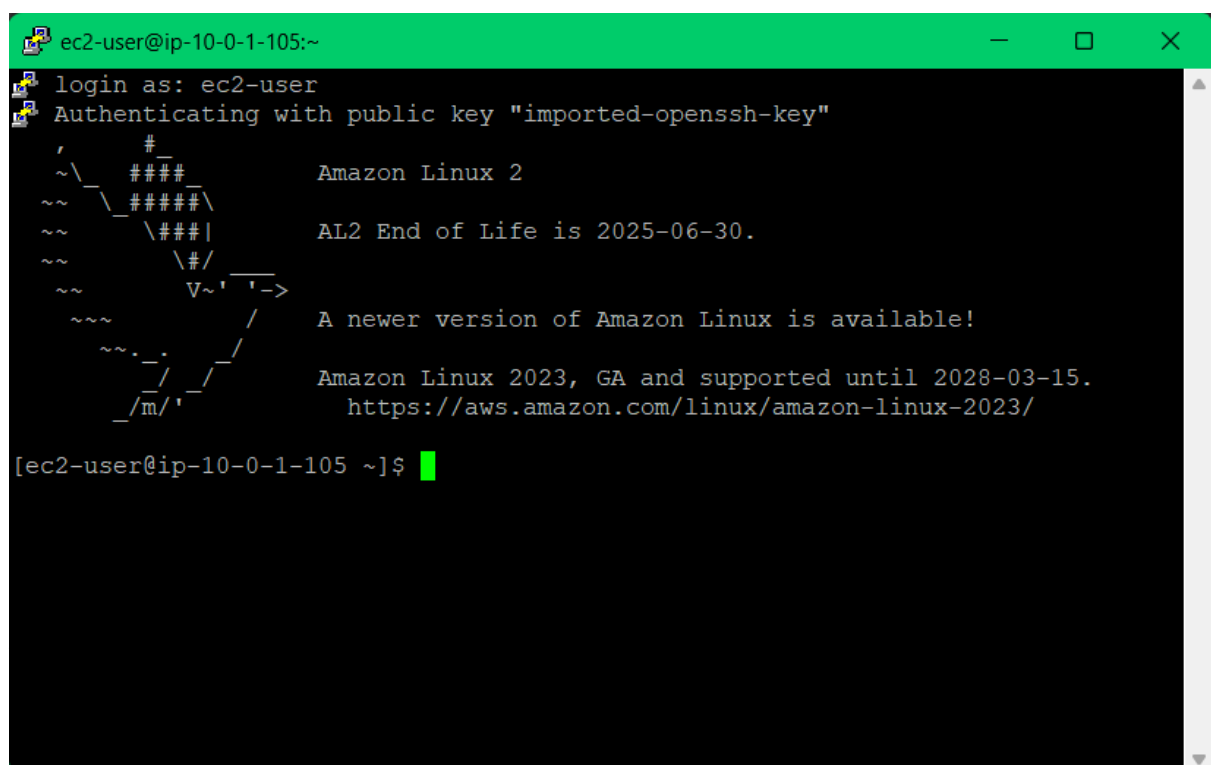
I copied the IPv4 Public IP from the Amazon EC2 console and paste it into PuTTY.

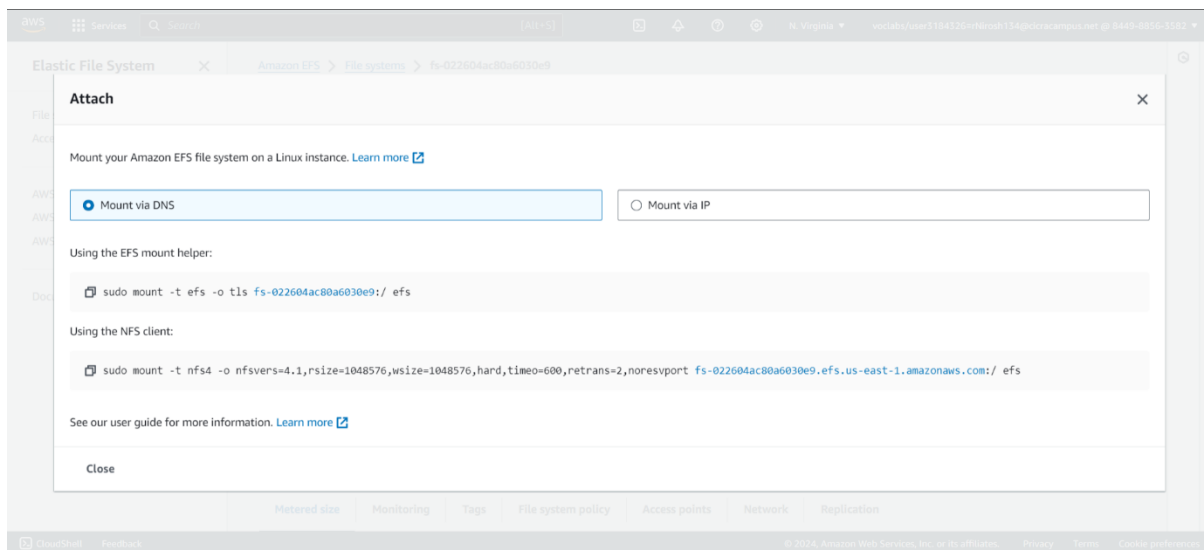


I navigate to PuTTY's SSH authentication settings and specify the labsuser.ppk file that I downloaded earlier.



I open the SSH connection to the EC2 instance. PuTTY prompts me to accept the host key to establish trust. Once connected, I'm prompted to log in as "ec2-user" to access the EC2 instance.

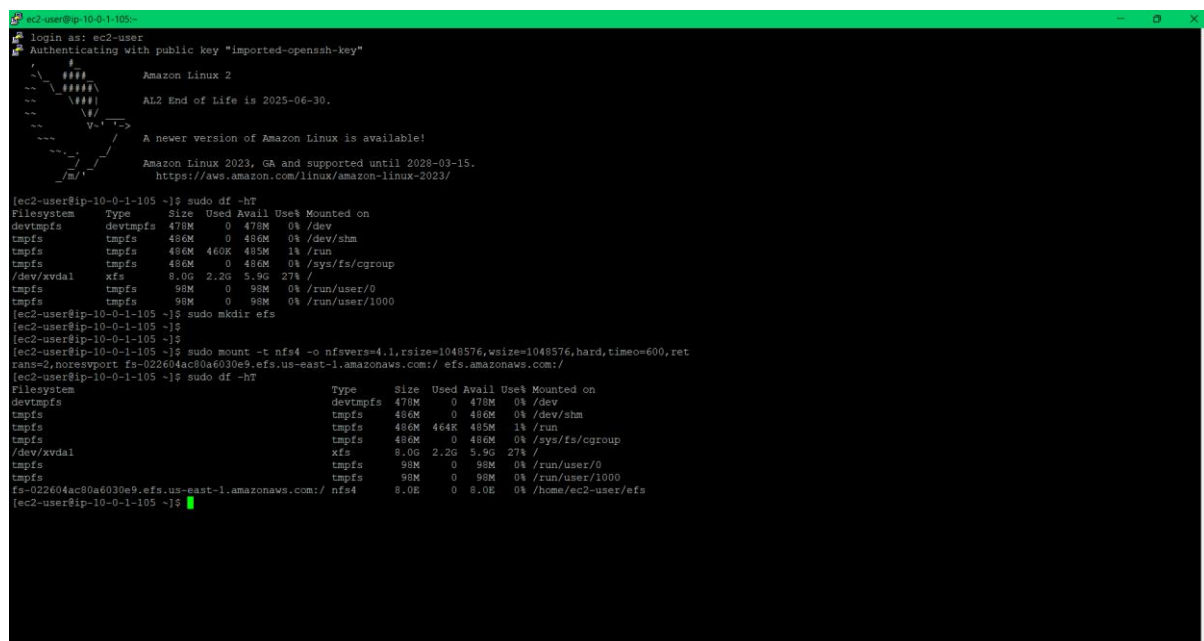




In my SSH session, I create a new directory named "efs" using the command `sudo mkdir efs`. I go back to the AWS Management Console and access the EFS service, there I select the specific EFS file system, such as "My First EFS File System". In the EFS Console, I choose "Attach" to open the instructions for mounting the EFS file system to an EC2 instance. I copy the entire mount command provided in the "Using the NFS client" section.

Back in my SSH session on the Linux EC2 instance, I paste the copied mount command and press ENTER to execute it. This mounts the EFS file system to the EC2 instance.

To get a summary of available and used disk space, I enter `sudo df -hT` in the SSH session. This command displays information about the mounted EFS file system, including its type and size.



Task 5: Examining the performance behavior of your new EFS file system

I just booted up my Amazon EC2 instance. A tool named fio is already installed. I'm going to run a test. The test should take around 5-10 minutes to complete. Once it's done, I'll take a look at the output to see the exact write speed.

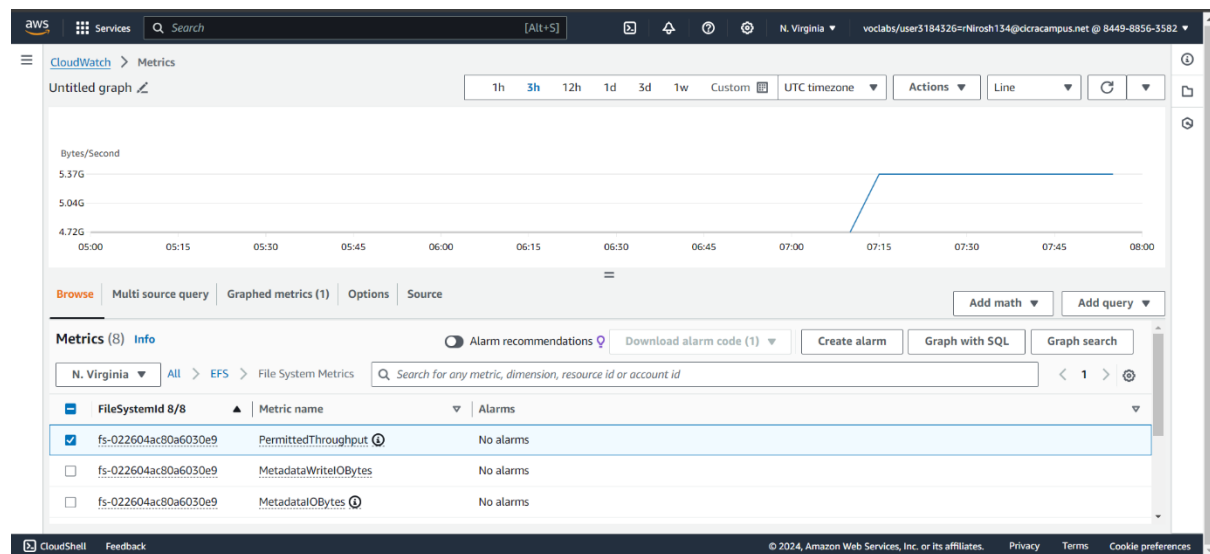
```
ec2-user@ip-10-0-1-105:~$ df -h
tmpfs      tmpfs      486M 460K 485M  1% /run
tmpfs      tmpfs      486M  0 486M  0% /sys/fs/cgroup
/dev/xvda1  xfs        8.0G 2.2G 5.9G 27% /
tmpfs      tmpfs      90M  0 90M  0% /run/user/0
tmpfs      tmpfs      98M  0 98M  0% /run/user/1000

[ec2-user@ip-10-0-1-105 ~]$ sudo mkdir /efs
[ec2-user@ip-10-0-1-105 ~]$
[ec2-user@ip-10-0-1-105 ~]$
[ec2-user@ip-10-0-1-105 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresport fs-022604ac80a6030e9.efs.us-east-1.amazonaws.com:/ /efs.amazonaws.com:/
[ec2-user@ip-10-0-1-105 ~]$ sudo df -hT
Filesystem                                Type      Size  Used Avail Use% Mounted on
devtmpfs                                  tmpfs     470M  0 470M  0% /dev
tmpfs                                     tmpfs     486M  0 486M  0% /dev/shm
tmpfs                                     tmpfs     486M 464K 485M  1% /run
tmpfs                                     tmpfs     486M  0 486M  0% /sys/fs/cgroup
/dev/xvda1                                xfs       8.0G 2.2G 5.9G 27% /
tmpfs                                     tmpfs     90M  0 90M  0% /run/user/0
tmpfs                                     tmpfs     98M  0 98M  0% /run/user/1000
fs-022604ac80a6030e9.efs.us-east-1.amazonaws.com:/ nfs4      8.0E  0 8.0E  0% /home/ec2-user/efs

[ec2-user@ip-10-0-1-105 ~]$ sudo fio --name=fio-efs --filesize=10G --filename=/efs/fio-efs-test.img --bs=1M --nrfiles=1 --direct=1 --sync=0 --rw=write --iodepth=200 --ioengine=libaio
fio-efs: (g=0): rw=write, bs=1M-1M/1M-1M, ioengine=libaio, iodepth=200
fio-2.14
Starting 1 process
fio-efs: Laying out 10 file(s) (1 file(s) / 10240MB)
Job: 1 (f=1): [W(1)] [98.8% done] [0KB/31744KB/0KB/s] [0/31/0 iops] [eta 00m:01s]
fio-efs: (groupid=0, jobs=1): err= 0: pid=32250: Tue Apr  9 08:12:58 2024
write: io=10240MB, bw=123854KB/s, iops=120, runt= 84662msec
slat (usec): min=53, max=290, avg=99.98, stdev=19.55
clat (msec): min=24, max=3411, avg=1652.87, stdev=198.54
lat (msec): min=24, max=3411, avg=1652.97, stdev=198.54
clat percentiles (msec):
 | 1.00th=[ 783], 5.00th=[ 1631], 10.00th=[ 1631], 20.00th=[ 1647],
 | 30.00th=[ 1647], 40.00th=[ 1647], 50.00th=[ 1647], 60.00th=[ 1663],
 | 70.00th=[ 1663], 80.00th=[ 1663], 90.00th=[ 1680], 95.00th=[ 1680],
 | 99.00th=[ 2507], 99.50th=[ 2933], 99.90th=[ 3294], 99.95th=[ 3326],
 | 99.99th=[ 3392]
lat (msec) : 50=0.03%, 100=0.12%, 250=0.21%, 500=0.31%, 750=0.29%
lat (msec) : 1000=0.30%, 2000=97.13%, >=2000=1.61%
cpu:
usr=0.70%, sys=0.61%, ctn=20678, majf=0, minf=1
IO depths : 1=0.1%, 2=0.1%, 4=0.1%, 8=0.1%, 16=0.2%, 32=0.3%, >=64=99.4%
submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.1%
issued    : total=r=0/w=10240/d=0, short=r=0/w=0/d=0, drop=r=0/w=0/d=0
latency   : target=0, window=0, percentile=100.00%, depth=200

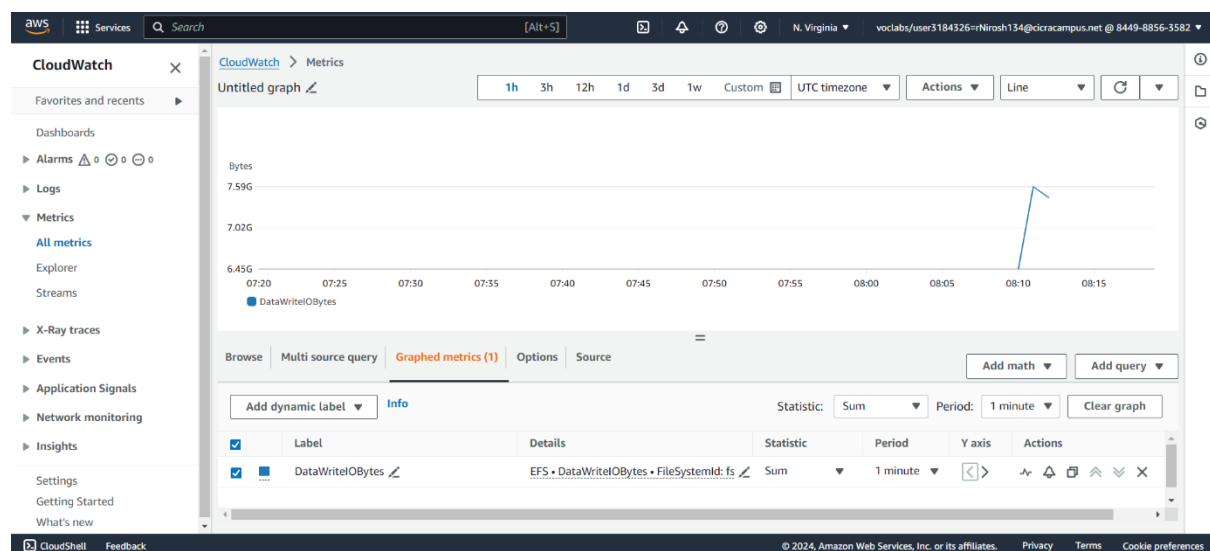
Run status group 0 (all jobs):
WRITE: io=10240MB, aggrbw=123854KB/s, minbw=123854KB/s, maxbw=123854KB/s, mint=84662msec, maxt=84662msec
[ec2-user@ip-10-0-1-105 ~]$
```

Monitoring performance by using Amazon CloudWatch



I navigated to the EFS metrics section and searched for a metric called "PermittedThroughput." It might take a few minutes for the data to populate.

After looking at the permitted throughput, I switched to another metric called "DataWriteIOBytes." This shows the actual write activity during a specific time period.



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Grade:

My Grade is 15/15

aws academy

ACAv2EN... > Assignments

> Module 4 Guided Lab - Introducing Amazon Elastic File System (Amazon EFS)

Module 4 Guided Lab - Introducing Amazon Elastic File System (Amazon EFS)

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab 0:22 Instructions Grades Actions

Files README Terminal Source

EN_US

Submitting your work

61. At the top of these instructions, choose **Submit** to record your progress and when prompted, choose **Yes**.

62. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this lab.

63. To find detailed feedback on your work, choose **Details** followed by **View Submission Report**.

```
3/asn2684771_1/tmp/temp_of_04892024/.38y928
mnt/vocwork3/grades/eee_0_2362885/asn2684770_3/asn2684771
_1/tmp/temp_of_04892024/.h008ft
Warning: Permanently added '54.225.134.86' (ECDSA) to the
of known hosts.
/home/ec2-user/efs/efs-test-ing
/usr/lib/python3.7/site-packages/boto3/compat.py:82: Pytho
nDeprecationWarning: Boto3 will no longer support Python 3.7 s
tarting December 13, 2023. To continue receiving service updat
es, bug fixes, and security updates please upgrade to Python 3
.8 or later. More information can be found here: https://aws.a
mazon.com/blogs/developer/python-support-policy-updates-for-au
s-sdks-and-tools/
warnings.warn(warning, PythonDeprecationWarning)
Present working directory = /mnt/vocwork3/grades/eee_0_236
2885/asn2684770_3/asn2684771_1/3184326/1/work
Default region: us-east-1
0
Back in submit.sh...
end
eee_N_3066282@punebl120802:~$
eee_N_3066282@punebl120802:~$
eee_N_3066282@punebl120802:~$
```

Total score 15/15

[Task 1] Security Group created	5/5
[Task 2] EFS file system created	5/5
[Task 5] Flexible IO was run	5/5